

Sphere

§ 1. Def. A sphere is the locus of a point which moves so that it is always at a constant distance from a fixed point. The fixed point is called the centre of the sphere and the distance is called the radius of the sphere.

The corresponding definition of the circle in plane geometry is identical with the definition of sphere.

§ 2. Equation of a sphere.

(a) Centre and radius being given. [Jiwaji 66]

If $P(x, y, z)$ be any point on the surface of a sphere whose centre is (a, b, c) and radius r , then by definition its equation is

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2 \quad \dots(1)$$

Thus the equation of a sphere whose centre is the point $(2, -3, 4)$ and radius 5 is $(x-2)^2 + (y+3)^2 + (z-4)^2 = 5^2$

or $x^2 + y^2 + z^2 - 4x + 6y - 8z + 4 = 0.$

[Punjab Hons. 52 ; Meerut 72]

The corresponding form of circle in plane geometry is

$$(x-h)^2 + (y-k)^2 = r^2.$$

Particular case—Centre being origin.

$$x^2 + y^2 + z^2 = r^2$$

(b) General form. [Putting $a=b=c=0$ in (1)] [Garhwal 79]

The equation $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$... (2) which is of second degree in x, y and z , (ii) in which the coefficient of x^2, y^2 and z^2 are equal, and (iii) which does not contain the terms of xy, yz, zx is the general equation of the sphere.

Note. If, however the coeff. of x^2, y^2 and z^2 be some constant p , we can divide throughout by p and reduce it to the above form.

The corresponding general form of the equation of a circle in plane geometry is $x^2 + y^2 + 2gx + 2fy + c = 0.$

(c) Centre and radius for general form.

The general equation (2) can be put as

$$(x^2 + 2ux + u^2) + (y^2 + 2vy + v^2) + (z^2 + 2wz + w^2) = -d \\ = (u^2 + v^2 + w^2) - d$$

or $(x+u)^2 + (y+v)^2 + (z+w)^2 = u^2 + v^2 + w^2 - d$

or $[x - (-u)]^2 + [y - (-v)]^2 + [z - (-w)]^2 = [\sqrt{(u^2 + v^2 + w^2 - d)}]^2$

i.e. centre is the point $(-u, -v, -w)$ and radius is

$$\sqrt{(u^2 + v^2 + w^2 - d)}.$$

Note. The coefficients of x^2, y^2, z^2 should be unity.

The corresponding result for circle is

centre $(-g, -f)$, radius $\sqrt{(g^2 + f^2 - c)}$.

~~Ex. 1. Find the centre and radius of the sphere whose equation is $ax^2 + ay^2 + az^2 + 2ux + 2vy + 2wz + d = 0$.~~ [Garhwal 79; Guru Nanak Dev 76]

Divide the given equation by a in order to reduce it to standard form in which the coefficient of x^2, y^2 and z^2 are each unity : we get

$$x^2 + y^2 + z^2 + \frac{2u}{a}x + \frac{2v}{a}y + \frac{2w}{a}z + \frac{d}{a} = 0$$

centre is $\left(-\frac{u}{a}, -\frac{v}{a}, -\frac{w}{a}\right)$

radius is $\left(\frac{u^2}{a^2} + \frac{v^2}{a^2} + \frac{w^2}{a^2} - \frac{d}{a}\right)^{1/2}$

$$= \frac{1}{a} (u^2 + v^2 + w^2 - ad)^{1/2} \quad \dots(1)$$

Another form

Prove that the equation

$$a(x^2 + y^2 + z^2) + 2(ux + vy + wz) + d = 0, a \neq 0$$

always represents a sphere, provided

$$u^2 + v^2 + w^2 - ad > 0$$

Since radius is to be real, the condition follows from (1).

(d) The equation of a sphere through four given points.

The general equation of a sphere contains four independent constants u, v, w and d and as such the equation of a sphere can be made to satisfy four conditions, e.g. can be made to pass through four given points.

General equation of a sphere is

$$(x^2 + y^2 + z^2) + 2ux + 2vy + 2wz + d = 0.$$

$\dots(1)$

Let the co-ordinates of the four given points be

$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$ and (x_4, y_4, z_4)

The points satisfy the above equation.

$$\therefore (x_1^2 + y_1^2 + z_1^2) + 2ux_1 + 2vy_1 + 2wz_1 + d = 0, \quad \dots(2)$$

$$(x_2^2 + y_2^2 + z_2^2) + 2ux_2 + 2vy_2 + 2wz_2 + d = 0, \quad \dots(3)$$

$$(x_3^2 + y_3^2 + z_3^2) + 2ux_3 + 2vy_3 + 2wz_3 + d = 0, \quad \dots(4)$$

$$(x_4^2 + y_4^2 + z_4^2) + 2ux_4 + 2vy_4 + 2wz_4 + d = 0, \quad \dots(5)$$

Eliminating the four constants, u, v, w and d from the above five equations, we get the required equation as

$x^2 + y^2 + z^2$	x	y	z	1
$x_1^2 + y_1^2 + z_1^2$	x_1	y_1	z_1	1
$x_2^2 + y_2^2 + z_2^2$	x_2	y_2	z_2	1
$x_3^2 + y_3^2 + z_3^2$	x_3	y_3	z_3	1
$x_4^2 + y_4^2 + z_4^2$	x_4	y_4	z_4	1

Ex. 2. Find the equation to the sphere which passes through the points $(0, 0, 0), (0, 1, -1), (-1, 2, 0)$ and $(1, 2, 3)$.

[Meerut 75 ; Madras 74]

Let the equation to the sphere be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0 \quad \dots(1)$$

Since it passes through origin $\therefore d = 0$.

Substitute the coordinates of other points we get

$$2 + 2v - 2w = 0 \quad \therefore v - w + 1 = 0 \quad \dots(2)$$

$$5 - 2u + 4v = 0 \quad \text{or} \quad 4v - 2u + 5 = 0 \quad \dots(3)$$

$$14 + 2u + 4v + 6w = 0 \quad \text{or} \quad u + 2v + 3w + 7 = 0 \quad \dots(4)$$

Eliminating w between (2) and (4), we get

$$u + 2v + 3(v - w + 1) + 7 = 0 \quad \therefore u + 5v + 10 = 0 \quad \dots(5)$$

Solving (3) and (5) for u and v we get

$$u = -\frac{1}{4}, v = -\frac{2}{11} \quad \text{hence from (2)} \quad w = -\frac{1}{11}$$

Putting for u, v, w and d the required equation is

$$x^2 + y^2 + z^2 - \frac{1}{4}x - \frac{2}{11}y - \frac{1}{11}z = 0$$

$$\text{or} \quad (x^2 + y^2 + z^2) - 15x - 25y - 11z = 0.$$

(b) Find the equation of the sphere which passes through the points $(1, -3, 4), (1, -5, 2)$ and $(1, -3, 0)$ and whose centre lies on the plane $x + y + z = 0$.

Let the equation of the sphere be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$$

[Jodhpur 78]

$$L = (x-x_1)(y-y_1)(z-z_1) + (x-x_2)(y-y_2)(z-z_2) + (x-x_3)(y-y_3)(z-z_3) + (x-x_4)(y-y_4)(z-z_4)$$

Co-ordinate Solid Geometry

It passes through the given points

$$\therefore 26 + 2u - 6v + 8w + d = 0 \quad \dots(1)$$

$$30 + 2u - 10v + 4w + d = 0 \quad \dots(2)$$

$$10 + 2u - 6v + 0w + d = 0 \quad \dots(3)$$

Subtracting in pairs we get

$$-4 + 4v + 4w = 0 \quad \text{or} \quad v + w = 1 \quad \dots(4)$$

$$20 - 4v + 4w = 0 \quad v - w = 5 \quad \dots(5)$$

Solving (4) and (5) we get $v = 3$, $w = -2$.

Again the centre $(-u, -v, -w)$ lies on the plane

$$x + y + z = 0 \quad \therefore -u - v - w = 0$$

$$u + v + w = 0 \quad \text{or} \quad u + 3 - 2 = 0 \quad \therefore u = -1.$$

Putting for u, v and w in any of (1), (2) or (3) we get $d = 10$.

Hence the required equation to the sphere is

$$x^2 + y^2 + z^2 - 2x + 6y - 4z + 10 = 0.$$

~~S = AL~~ ~~(4)~~ Find the centre of the sphere through the four points
(4, 1, 2) (0, -2, 3), (1, -5, 1) and (2, 0, 1). [Berhampur 83]

~~Ruhman~~ Ans. (2, -13/5, 7/5).

(e) Equation of a sphere on the join of two given points as diameter. [Berhampur 81; Cal. 76; Gaubati 74; Luck. 78; Indore 82; Andhra 84; M.D. 79; Magadh Hons. 68]

Let the co-ordinates of the given points A and B be (x_1, y_1, z_1) , and (x_2, y_2, z_2) respectively and $P(x, y, z)$ be any point on the sphere. The angle subtended at P by the line AB is a right angle (angle in a semi-circle is a right angle), i.e. the line AP is perpendicular to line BP .

The direction-ratios of AP are $x - x_1$, $y - y_1$, $z - z_1$ and those of BP are $x - x_2$, $y - y_2$, $z - z_2$. Since these two lines are perpendicular, we have

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0 \quad \dots(6)$$

Above is the required equation.

Ex. 3. Find the equation of the sphere on the join of (2, -3, 4) and (-5, 6, -7) as diameter. [Gorukhpur 76]

$$(x - 2)(x + 5) + (y + 3)(y - 6) + (z - 4)(z + 7) = 0$$

$$x^2 + y^2 + z^2 + 3x - 3y + 3z - 56 = 0$$

or

Exercise 19

~~Ex 1.~~ Find the equation of a sphere which passes through origin and intercepts lengths a, b and c on the axes respectively.

[Ranchi 76; Agra 74; Gujrat 65]

Or

Find the equation of a sphere passing through origin and the points, $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$. [Osmania 83 ; V. Imp]

Or

If the plane $x/a + y/b + z/c = 1$ meets the coordinate axes in points A, B, C and O be the origin. Find the sphere $OABC$.

[Raj. 78 ; Gorakhpur 73]

Let the equation of sphere $OABC$ be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0.$$

It passes through origin; $\therefore d = 0$.

Hence its equation is

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0.$$

Since the sphere passes through the point $A (a, 0, 0)$ and similarly through $B (0, b, 0)$ and $C (0, 0, c)$.

It passes through $A (a, 0, 0)$; $\therefore a^2 + 2ua = 0$ or $2u = -a$.
Similarly $2v = -b$ and $2w = -c$.

Hence the equation of sphere $OABC$ is

$$x^2 + y^2 + z^2 - ax - by - cz = 0.$$

[Remember]

Ex. 2. Find the equation of the sphere circumscribing the tetrahedron whose faces are

$$x = 0, y = 0, z = 0, \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

[Meerut 80 ; Punjab 76]

Solving and taking three planes at a time, we get the four vertices as $O (0, 0, 0)$, $A (a, 0, 0)$ and $B (0, b, 0)$ and $C (0, 0, c)$, so that the sphere $OABC$ as in Ex. 1. is

$$x^2 + y^2 + z^2 - ax - by - cz = 0.$$

Ex. 3. Find the equation of a sphere which passes through origin and makes equal intercepts of unit length on the axes.

Ans. $x^2 + y^2 + z^2 - x - y - z = 0$.

Ex. 4. (a) A sphere of radius k passes through the origin and

meets the axes in A, B, C . Prove that the centroid of the triangle ABC lies on the sphere, $9(x^2 + y^2 + z^2) = 4k^2$.

[Rewa 74 ; Avadh 82, Vikram 72 ; Gauhati Hons. 73 ;

Nagpur 77 ; Kanpur 79 ; Rajasthan 84 ; Allahabad 80 ;

Madurai 83 ; Burdwan Hons. 83 ; Indore 78]

Let the co-ordinates of A, B and C be $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$ and the fourth point be $(0, 0, 0)$. The equation of the sphere through these points is

$$x^2 + y^2 + z^2 - ax - by - cz = 0,$$

$$\text{It radius is } \sqrt{\left(\frac{a^2}{4} + \frac{b^2}{4} + \frac{c^2}{4}\right) = k^2},$$

$$\therefore a^2 + b^2 + c^2 = 4k^2. \quad \dots(1)$$

If (f, g, h) be the co-ordinates of the centroid of $\triangle ABC$ then $f = a/3$, $g = b/3$ and $h = c/3$. Putting in (1) for a, b, c , we get $9(f^2 + g^2 + h^2) = 4k^2$. Generalising, we get the required locus.

(b) A sphere of constant radius $2k$ passes through origin and meets the axes in A, B, C . Find the locus of the centroid of the tetrahedron $OABC$.

Sphere $OABC$ is

$$x^2 + y^2 + z^2 - ax - by - cz = 0. \quad (\text{by Ex. 1})$$

$$\text{where } \frac{a^2}{4} + \frac{b^2}{4} + \frac{c^2}{4} = 4k^2. \quad \dots(1)$$

If (f, g, h) be C.G. of tetrahedron $OABC$, then

$$f = \frac{0+a+0+0}{4}, \therefore a=4f. \text{ similarly, } b=4g \text{ and } c=4h.$$

Hence from (1), we get

$$16(f^2 + g^2 + h^2) = 16k^2 \quad \text{or} \quad f^2 + g^2 + h^2 = k^2.$$

Generalizing, the locus is $x^2 + y^2 + z^2 = k^2$.

Ex. 5. Find the equation of the sphere which passes through the points $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ and radius as small as possible. [Raj. 77 ; Indore 76 ; Agra 81]

Let the sphere be

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0.$$

It passes through $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$.

$$2u = 2v = 2w = -(1+d).$$

If r be the radius, then $r^2 = u^2 + v^2 + w^2 - d$. Now let $z = r^2$.

$$\therefore z = r^2 = \frac{3}{4}(1+d)^2 - d. \quad \text{On putting for } u, v \text{ and } w.$$

If r is least, then r^2 , i.e. z is least

$$\therefore \frac{dz}{dd} = 0 \text{ or } \frac{3}{2} \cdot 2(1+d) - 1 = 0; \quad d = -\frac{1}{3}.$$

This value of d makes $\frac{d^2z}{dd^2}$ +ive and hence least.

$$\therefore 2u = 2v = 2w = -(1 - \frac{1}{3}) = -\frac{2}{3}.$$

Hence the equation of sphere is

$$\begin{aligned} x^2 + y^2 + z^2 - \frac{2}{3}(x+y+z) - \frac{1}{3} &= 0. \\ \text{or } 3(x^2 + y^2 + z^2) - 2(x+y+z) - 1 &= 0. \end{aligned}$$

Ex. 6. A plane passes through a fixed point (p, q, r) and cuts the axes in A, B, C . Show that the locus of the centre of the sphere $OABC$ is $\frac{p}{x} + \frac{q}{y} + \frac{r}{z} = 2$.

[Grahwal 75; Madras 75; Sagar 73; Indore 83; Utkal Hons. 81; Rohailkhand 80; Ravishanker 81; Jiwaji 81; Ranchi 83, 78; Karnatak 83; Calicut 84; Rajputana 83; Meerut 83; Kanpur 82; Allahabad 81]

Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

and hence the points A, B, C are $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$ and the equation of the sphere $OABC$ is $x^2 + y^2 + z^2 - ax - by - cz = 0$. If (f, g, h) be the centre, then

$$f = \frac{a}{2}, g = \frac{b}{2} \text{ and } h = \frac{c}{2} \quad \dots(1)$$

Now the plane (1) passes through (p, q, r) .

$$\therefore p/a + q/b + r/c = 1. \quad \dots(2)$$

Putting the values of a, b and c from (2) in (3), we get $p/2f + q/2g + r/2h = 1$. Generalizing, (f, g, h) we get the locus as $p/x + q/y + r/z = 2$. $\dots(3)$

Ex. 7. A sphere of constant radius r passes through the origin O and cuts the axes in A, B and C . Prove that the locus of the foot of the perpendicular from O to the plane ABC is given by $(x^2 + y^2 + z^2)^2 (x^{-2} + y^{-2} + z^{-2}) = 4r^2$.

[Rajasthan 79, 81, 84; Rohailkhand 81, 79 Meerut 73; Kanpur 83; Delhi 76; Allahabad 79, 75; Nagpur 78; Osmania 82; Vikram 66]

Plane is

$$x/a + y/b + z/c = 1.$$

Sphere is

$$x^2 + y^2 + z^2 - ax - by - cz = 0,$$

where

$$a^2 + b^2 + c^2 = 4r^2. \quad \dots(1)$$

Any line through origin perpendicular to plane (1) is

$$\frac{x}{1/a} = \frac{y}{1/b} = \frac{z}{1/c} = \lambda.$$

Any point on it is $(\lambda/a, \lambda/b, \lambda/c)$.

If it is foot of perpendicular (x, y, z) , then

$$x = \frac{\lambda}{a}, y = \frac{\lambda}{b}, z = \frac{\lambda}{c}; \therefore a = \frac{\lambda}{x}, b = \frac{\lambda}{y} \text{ and } c = \frac{\lambda}{z}.$$

Putting in (3), we get

$$\lambda^2 (x^{-2} + y^{-2} + z^{-2}) = 4r^2. \quad \dots(4)$$

Putting in (1), we get $\frac{1}{\lambda} (x^2 + y^2 + z^2) = 1$.

Eliminating λ between (4) and (5), we get the required locus as $(x^2 + y^2 + z^2)^2 (x^2 + y^2 + z^2) = 4r^2$ (5)

Ex. 8. OA , OB and OC are mutually perpendicular lines through the origin and their direction-cosines are

$$l_1, m_1, n_1; l_2, m_2, n_2; l_3, m_3, n_3.$$

If $OA=a$, $OB=b$, $OC=c$, prove that the equation of the sphere $OABC$ is

$$x^2 + y^2 + z^2 - x(al_1 + bl_2 + cl_3) - y(am_1 + bm_2 + cm_3) - z(an_1 + bn_2 + cn_3) = 0.$$

[Burdwan Hons. 75; Pb. 52 Agra 64]

The co-ordinates of the points O , A , B and C are respectively. (§ 3 P. 17)

$$(0, 0, 0), (al_1, am_1, an_1), (bl_2, bm_2, bn_2), (cl_3, cm_3, cn_3).$$

Since the sphere passes through origin, its equation can be put as $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$ (1)

Substitute the co-ordinates of the other three points.

$$\therefore a + 2ul_1 + 2vm_1 + 2wn_1 = 0 \quad \dots (2)$$

$$\therefore \Sigma l_1^2 = l_1^2 + m_1^2 + n_1^2 = 1. \quad \dots (3)$$

$$\text{Similarly } b + 2ul_2 + 2vm_2 + 2wn_2 = 0, \quad \dots (4)$$

$$c + 2ul_3 + 2vm_3 + 2wn_3 = 0. \quad \dots (5)$$

Now l_1, l_2, l_3 are the direction-cosines of OX with reference to the three mutually perpendicular given lines.

$$\therefore l_1^2 + l_2^2 + l_3^2 = m_1^2 + m_2^2 + m_3^2 = n_1^2 + n_2^2 + n_3^2 = 1$$

$$l_1m_1 + l_2m_2 + l_3m_3 = 0 = m_1n_1 + m_2n_2 + m_3n_3 = n_1l_1 + n_2l_2 + n_3l_3.$$

Multiplying (2), (3) and (4) by l_1, l_2 and l_3 respectively and using the above relations, we get

$$(al_1 + bl_2 + cl_3) + 2u = 0. \therefore 2u = -(al_1 + bl_2 + cl_3).$$

Similarly multiplying (2), (3), (4) by m_1, m_2 and m_3 and then by n_1, n_2, n_3 we get

$$2v = -(am_1 + bm_2 + cm_3), \quad 2w = -(an_1 + bn_2 + cn_3).$$

Putting the values of $2u, 2v$ and $2w$ in (1), we get the equation of the sphere as given.

Ex. 9. Find the equation of a sphere through the four points $(0, 0, 0)$, $(-a, b, c)$, $(a, -b, c)$, $(a, b, -c)$ and determine its radius.

[Venka 84; Cal. 83]

Since the sphere passes through origin, its equation is of the form $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$ (1)

Substituting the co-ordinates of the other given points, we

$$a^3 + b^3 + c^3 - 2ua - 2vb + 2wc = 0. \quad \dots(2)$$

$$a^3 + b^3 + c^3 + 2ua - 2vb + 2wc = 0. \quad \dots(3)$$

$$a^3 + b^3 + c^3 + 2ua + 2vb - 2wc = 0. \quad \dots(4)$$

Adding (2) and (3), $2(a^3 + b^3 + c^3) = -4wc$ or $2w = \frac{a^3 + b^3 + c^3}{-c}$.

Similarly, $2v = \frac{a^3 + b^3 + c^3}{-c}$ and $2u = \frac{a^3 + b^3 + c^3}{-a}$.

Putting the values of u, v, w in (1), we get

$$x^3 + y^3 + z^3 - (a^3 + b^3 + c^3) \left(\frac{x}{a} + \frac{y}{b} + \frac{z}{c} \right) = 0.$$

The radius of the sphere is $\sqrt{(u^2 + v^2 + w^2)}$, d being zero,

$$\begin{aligned} \text{or } R &= \sqrt{\left\{ \frac{(a^3 + b^3 + c^3)^2}{4} \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) \right\}} \\ &= \frac{a^3 + b^3 + c^3}{2} \sqrt{(a^{-2} + b^{-2} + c^{-2})}. \end{aligned}$$

Another form. Prove that the equation to a sphere circumscribing the tetrahedron whose faces are

$$\frac{y}{b} + \frac{z}{c} = 0, \quad \frac{z}{c} + \frac{x}{a} = 0, \quad \frac{x}{a} + \frac{y}{b} = 0, \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\text{is } \frac{x^3 + y^3 + z^3}{a^3 + b^3 + c^3} - \frac{x}{a} - \frac{y}{b} - \frac{z}{c} = 0.$$

[Vikram 64]

It we solve the four equations taking three at a time, we get the points $(0, 0, 0)$, $(-a, b, c)$, $(a, -b, c)$, $(a, b, -c)$.

Now see last Ex. [9].

(b) Find the equation of the sphere circumscribing the tetrahedron formed by the planes

$$4y + 3z = 0; \quad 3x + 2y = 0; \quad z + 2x = 0 \text{ and } 6x + 4y + 3z = 12$$

[Burdwan 74]

The first three meet at $(0, 0, 0)$.

Now solving 1st, 2nd and 4th we get

$$6x + 0 = 12, \quad \therefore x = 2, \quad \therefore y = -3 \text{ and hence } z = 0$$

$\therefore$ vertex is $(2, -3, 4)$.

Similarly other vertices are $(-2, 3, 4)$ and $(2, 3, -4)$

Let the sphere be $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$.

Since it passes through $(0, 0, 0) \quad \therefore d = 0$

It passes through other three points

$$\therefore \begin{aligned} 29+4u-6v+8w &= 0 \\ 29-4u+6v+8w &= 0 \\ 29+4u+6v-8w &= 0 \end{aligned}$$

Adding in pairs $29(2)+16w=0 \quad \therefore 2w=-\frac{29}{4}$

Similarly $2v=-29/3, 2u=-29/2$

Hence the required sphere is

$$x^2+y^2+z^2-29\left(\frac{x}{2}+\frac{y}{3}+\frac{z}{4}\right)=0$$

or $\frac{x^2+y^2+z^2}{2^2+3^2+4^2}=\frac{x}{2}+\frac{y}{3}+\frac{z}{4}$

or $12(x^2+y^2+z^2)=174x+116y+87z$

Ex. 10. Determine the equation of the sphere having its centre on the line $5y+2z=0=2x-3y$ and passing through the points $(0, -2, -4)$ and $(2, -1, -1)$. [Agra 77; Lucknow 81]

Let the equation of the sphere be

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0 \quad \dots(1)$$

Centre $(-u, -v, -w)$ lies on the given line

$$\therefore 5v+2w=0 \quad \dots(2) \text{ and } 2u-3v=0 \quad \dots(3)$$

It passes through given points

$$\therefore 20-4v-8w+d=0 \quad \dots(4)$$

$$6+4u-2v-2w+d=0 \quad \dots(5)$$

Subtracting (4) and (5) we get

$$14-4u-2v-6w=0 \text{ or } 2u+v+3w=7 \quad \dots(6)$$

Putting for u and w from (2) and (3) in (6) we get

$$v=\frac{2}{3}u, \quad w=-\frac{5}{2}v=-\frac{5}{2}\cdot\frac{2}{3}u=-\frac{5}{3}u$$

$$\therefore 2u+\frac{2}{3}u+3\left(-\frac{5}{3}u\right)=7$$

$$\text{or } (6+2-15)u=21 \quad \therefore u=-3 \text{ and } v=-2, w=-5$$

Putting for u, v and w in (4) or (5) we get $d=12$.

Hence the equation of the sphere is

$$x^2+y^2+z^2-6x-4y+10z+12=0$$

Ex. 11. (a) Find the equation of the sphere that passes through the points $(4, 1, 0)$, $(2, -3, 4)$, $(1, 0, 0)$ and touch the plane $2x+2y-z=11$. [Garhwal 78; Avadh 82]

Let the equation of the sphere be

$$x^2+y^2+z^2+2ux+2vy+2wz+d=0 \quad \dots(1)$$

It passes through the given points

$$17+8u+2v+d=0 \quad \dots(2)$$

$$29+4u-6v+8w+d=0 \quad \dots(3)$$

$$1+2u+d=0 \quad \dots(4)$$

The sphere touches the given plane and hence perpendicular from centre $(-u, -v, -w)$ is equal to radius $\sqrt{(u^2 + v^2 + w^2 - d)}$

$$\therefore \frac{-2u - 2v + w - 11}{\sqrt{(4 + 4 + 1)}} = \sqrt{(u^2 + v^2 + w^2 - d)}$$

$$\therefore (2u + 2v - w + 11)^2 = 9(u^2 + v^2 + w^2 - d)$$

$$(2u + 2v - w)^2 + 22(2u + 2v - w) + 121$$

$$= 9(u^2 + v^2 + w^2 - d)$$

$$\text{or } 5u^2 + 5v^2 + 8w^2 - 8uv + 4vw + 4uw$$

$$- 44u - 44v + 22w - 9d - 121 = 0$$

...(5)

$$\text{Now from (4) } 4 = -\frac{1}{2}(d + 1)$$

$$\text{Putting for } u \text{ in (2) we get } v = \frac{1}{2}(3d - 13)$$

$$\text{Finally putting for } u \text{ and } v \text{ in (3) we get } w = \frac{1}{2}(5d - 33)$$

Now putting the values of u, v, w in (5) we get on simplification

$$72d^2 - 747d + 1935 = 0$$

Clearly $d = 5$ satisfies it

$$\text{Putting for } d \text{ we get } u = -3, v = 1, w = -2$$

Hence the equation to the sphere is

$$x^2 + y^2 + z^2 - 6x + 2y - 4z + 5 = 0$$

(b) Obtain the equation of the sphere with centre at $(1, -2, 3)$ and touching the plane $6x - 3y + 2z = 4$.

Since the sphere touches the given plane therefore perpendicular from centre is equal to radius

$$\therefore \frac{6(1) - 3(-2) + 2(3) - 4}{\sqrt{(36 + 9 + 4)}} = r \quad \text{or} \quad \frac{14}{7} = r = 2.$$

$$\text{Hence the equation is } (x - 1)^2 + (y + 2)^2 + (z - 3)^2 = 2^2$$

$$\text{or } x^2 + y^2 + z^2 - 2x + 4y - 6z + 10 = 0$$

Ex. 12. Find the equation of the sphere on the join of $(2, -3, 1)$ and $(3, -1, 2)$ as diameter.

Using (e) P. 199, the required equation is

$$(x - 2)(x - 3) + (y + 3)(y + 1) + (z - 1)(z - 2) = 0$$

$$\text{or } x^2 + y^2 + z^2 - 5x + 4y - 3z + 11 = 0.$$

$$\text{Centre is } (-u, -v, -w) \text{ i.e. } \left\{\frac{5}{2}, -2, \frac{3}{2}\right\}$$

$$\text{radius} = \sqrt{(u^2 + v^2 + w^2 - d)} = \frac{1}{2}\sqrt{6}.$$

Ex. 13. A point moves such that sum of the squares of its distances from a given number of points is constant. Prove that the locus of this point is a sphere, show that its centre is centroid of the given points.

$$\Sigma[(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2] = r^2$$

$$\text{or } nx^2 + ny^2 + nz^2 - 2x\Sigma x_1 - 2y\Sigma y_1 - 2z\Sigma z_1 + \Sigma x_1^2$$

$$+ \Sigma y_1^2 + \Sigma z_1^2 = 0.$$

Dividing by n , we get the co-ordinates of the centre as $\left(\frac{\Sigma x_1}{n}, \frac{\Sigma y_1}{n}, \frac{\Sigma z_1}{n}\right)$ which is the centroid of given points.

Ex. 14. A point moves so that the sum of the squares of its distances from the six faces of a cube is constant, show that its locus is a sphere. [Meerut 79]

Refer fig. 4. P. 3. If a be the length of an edge of a cube then the six planes in pairs of parallel planes are

$$x=0, x=a, y=0, y=a, z=0, z=a.$$

Let (f, g, h) be a moving point then by the given condition

$$\left(\frac{f}{1}\right)^2 + \left(\frac{f-a}{1}\right)^2 + \left(\frac{g}{1}\right)^2 + \left(\frac{g-a}{1}\right)^2 + \left(\frac{h}{1}\right)^2 + \left(\frac{h-a}{1}\right)^2 = k^2$$

$$\text{or } 2(f^2 + g^2 + h^2) - 2(fa + ga + ha) + 3a^2 - k^2 = 0$$

$$\text{or } f^2 + g^2 + h^2 - (fa + ga + ha) + \frac{1}{2}(3a^2 - k^2) = 0.$$

Hence the locus of the point (f, g, h) is

$$x^2 + y^2 + z^2 - a(x + y + z) + d = 0$$

$$\text{where } d = \frac{1}{2}(3a^2 - k^2)$$

Above equation represents a sphere.

Ex. 15. A plane passes through a fixed point (a, b, c) show that the locus of foot of the perpendicular to it from the origin is the sphere $OABC$

$$x^2 + y^2 + z^2 - ax - by - cz = 0.$$

Any plane through (a, b, c) is

$$l(x-a) + m(y-b) + n(z-c) = 0. \quad \dots(1)$$

Any line through origin and perpendicular to it i.e. parallel to its normal l, m, n is

$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n}. \quad \dots(2)$$

(1) and (2) intersect in foot of perpendicular whose locus is obtained by eliminating the variables l, m, n between (1) and (2) and is

$$x(x-a) + y(y-b) + z(z-c) = 0$$

$$\text{or } x^2 + y^2 + z^2 - ax - by - cz = 0, \text{ i.e. sphere } OABC.$$

§ 3. Plane Section of a sphere.

To prove that the section of a sphere by a plane is a circle and to find its radius and centre. [G.N.D. 73; Punjab 63]

The dotted line denotes the section of the sphere by a plane and let P be any point on it. From O draw ON perpendicular to the plane and the foot of the perpendicular be N . Now NP is any line in the plane and ON is perpendicular to the plane; thus ON is perpendicular to every line lying in the plane.

$\therefore ON$ is $\perp$ to NP and as such

$$OP^2 = ON^2 + NP^2$$

or $NP = \sqrt{(OP^2 - ON^2)}$ = constant as OP is equal to the radius of the sphere and ON is the perpendicular from the centre on the given plane.

Thus for all positions of P , NP is constant hence the locus of this point P is a circle whose centre is the point N and radius NP .

Centre. The foot of the perpendicular from the centre of the sphere to the plane is the centre of the circle.

$$(\text{radius})^2 = (\text{radius of sphere})^2 - (\perp \text{ from centre on the plane})^2.$$

Great Circle. The section of a sphere by a plane through the centre of the sphere is a great circle. Its centre and radius are the same as those of the given sphere.

§ 4. Curve of intersection of two spheres. To prove that the curve of intersection of two spheres is a circle. [Raj. 75]

Let $S_1 \equiv 0$, $S_2 \equiv 0$ be the equations of two spheres with coefficients of x^2 , y^2 and z^2 each unity in both. Now the points of intersection of S_1 and S_2 i.e. the points for which $S_1 \equiv 0$, $S_2 \equiv 0$, also satisfy the equation $S_1 - S_2 \equiv 0$; but this equation will reduce to a first degree equation and therefore represents a plane. Thus the points of intersection of two spheres are the same as those of the intersection of any of the spheres and the plane $S_1 - S_2 \equiv 0$ and thus these points lie on a circle.

§ 5. Sphere through a given circle.

[Utkal 69; Gaubati 66; Sagar 63; Vikram 61]

From § 3 and § 4 we conclude that the equation of a sphere and plane together represents a circle or two equations of spheres taken together represents a circle.

$$\text{Let } S \equiv x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + k$$

and

$$P \equiv ax + by + cz + d.$$

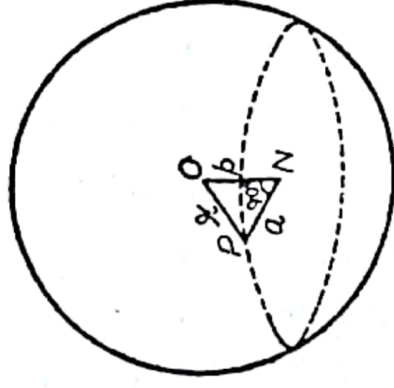


FIG-34

Now Let $S=0$ be the equation of a sphere and $P=0$ that of a plane; then the equation $S+\lambda P=0$ for all values of λ is satisfied by all those points which satisfy both $S=0$ and $P=0$, i.e. those which lie on the circle $S=0$ and $P=0$. The equation $S+\lambda P=0$ being of 2nd degree, having the coefficients of x^2, y^2, z^2 equal and not containing the terms of xy, yz and zx , represents a sphere through the circle whose equations are $S=0, P=0$.

Similarly $S+\lambda S'=0$ also represents a sphere through the circle whose equations are $S=0$ and $S'=0$.

The value of λ is found by an additional given condition.

Again we know that a circle can be made to pass through any three points. Thus if a sphere is to contain a given circle it should pass through three points.

Note. The equation $x^2+y^2+2gx+2fy+c=0, z=0$ also represents a circle in $x-y$ plane.

Exercise 20

Ex. 1. (a) Find the co-ordinates of the centre and the radius of the circle $x+2y+2z=15, x^2+y^2+z^2-2y-4z=11$.

[Agra 76; Gauhati 74; Sambalpur 74; Madras 74; B. khand 78]

Length of the perpendicular from the centre $(0, 1, 2)$ on the

$$\text{plane is } \frac{0+2+4-15}{\sqrt{(1+4+4)}} = \frac{-9}{3} = -3 = r.$$

$$\therefore (\text{radius of circle})^2 = (\text{radius of sphere})^2 - (\text{perp. from centre on the plane})^2 = 16-9=7; \therefore r=\sqrt{7}.$$

Centre of the circle is the foot of the perpendicular from the centre of the sphere $(0, 1, 2)$ to the plane $x+2y+2z=15$. Any line through $(0, 1, 2)$ perpendicular to plane, i.e. parallel to its normal whose D.R.'s are $-1, 2, 2$ is

$$\frac{x-0}{1} = \frac{y-1}{2} = \frac{z-2}{2} = r \text{ say.}$$

Ans.

Any point on it is $r, 2r+1, 2r+2$. For foot of perpendicular this must lie on the given plane;

$$\therefore r+2(2r+1)+2(2r+2)=15 \text{ or } 9r=9; \therefore r=1.$$

Hence the centre of circle is $(1, 3, 4)$.

Ex. 1. (b) Find the centre and radius of the circle

$$x^2+y^2+z^2=25, x+2y+2z+9=0.$$

[Burdwan 87]

Ans. Radius = 4, centre is $(-1, -2, -2)$.

Ex. 2. Obtain the equation of the circle lying on the sphere $x^2+y^2+z^2-2x+4y-6z+3=0$ and having its centre at $(2, 3, -4)$.

The required equations are the equations of the given sphere and of a plane through $(2, 3, -4)$ and whose normal is the line joining $(2, 3, -4)$ to the centre of the sphere i.e. $(1, -2, 3)$, $\therefore$ D.R.'s of its normal are $x_2 - x_1, y_2 - y_1, z_2 - z_1$ i.e. $1, 5, -7$ and hence its equation is

$$1(x-2) + 5(y-3) - 7(z+4) = 0$$

Hence the equations of the circle are given by

$$P=0 \text{ and } S=0$$

$$\text{or } x + 5y - 7z - 45 = 0 \text{ and } x^2 + y^2 + z^2 - 2x + 4y - 6z + 3 = 0.$$

(b) Find the equation of the circle whose centre is (x_1, y_1, z_1) and which lies on the sphere $x^2 + y^2 + z^2 = a^2$.

Proceed as above. The required equations are

$$P \equiv x_1(x-x_1) + y_1(y-y_1) + z_1(z-z_1) = 0$$

$$S \equiv x^2 + y^2 + z^2 - a^2 = 0.$$

(c) Prove that the centre of all sections of the sphere $x^2 + y^2 + z^2 = r^2$ by planes through point (x', y', z') lie on the sphere $x(x-x') + y(y-y') + z(z-z') = 0$. [Madras 76]

Centre of the given sphere is $(0, 0, 0)$. If (f, g, h) be the centre of the section, then the line joining $(0, 0, 0)$ and (f, g, h) is normal to the plane and its direction-ratios are (f, g, h) . Hence the equation of the plane cutting the sphere in a circle with centre (f, g, h) is

$$f(x-f) + g(y-g) + h(z-h) = 0. \quad \dots(1)$$

If it passes through the point (x', y', z') , we have

$$f(x'-f) + g(y'-g) + h(z'-h) = 0$$

Hence the locus of the point (f, g, h) is as given

$$-x(x-x') + y(y-y') + z(z-z') = 0.$$

Ex. 3. Find the equation of the sphere through the circle $x^2 + y^2 + z^2 = 9$, $2x + 3y + 4z = 5$ and the point $(1, 2, 3)$. [Agra 82; Osmania 82; Calicut 84; Sambalpur 73; Gorakhpur 81; Gauhati 78]

Sphere through the given circle is given by $S + \lambda P = 0$.

$$\text{or } (x^2 + y^2 + z^2 - 9) + \lambda(2x + 3y + 4z - 5) = 0.$$

$$\text{It passes through } (1, 2, 3); \therefore \lambda = -\frac{1}{3}.$$

$\therefore$ Required sphere is

$$3(x^2 + y^2 + z^2) - 2x - 3y - 4z - 22 = 0.$$

(b) Find the equation of the sphere through the circle $x^2 + y^2 = a^2$, $z = 0$ and through the point (α, β, γ) . [Meerut 83, 71; Agra 80; Gorakhpur 82; Kanpur 77; Punjab 78]

Since $z=0$ the equation of the circle can be written as

$$S = x^2 + y^2 + z^2 = a^2 \text{ and } P = z = 0. \quad (\text{Note } z^2)$$

The sphere through the given circle is $S + \lambda P = 0$

Pass through (α, β, γ) find the value of λ .

$$\text{Ans. } (x^2 + y^2 + z^2 - a^2) \gamma + (a^2 - \alpha^2 - \beta^2 - \gamma^2) z = 0.$$

(c) A circle with centre $(2, 3, 0)$ and radius 1 is drawn in the plane $z=0$. Find the equation of the sphere which passes through this circle and the point $(1, 1, 1)$. [Luck. 79]

$$\text{Circle is } (x-2)^2 + (y-3)^2 = 1, z=0$$

$$\text{Write it as } S = (x-2)^2 + (y-3)^2 + z^2 - 1 = 0, P = z = 0$$

Required sphere is $S + \lambda P = 0$ and pass through $(1, 1, 1)$ so that $\lambda = -5$.

$$\text{Ans. } x^2 + y^2 + z^2 - 4x - 6y - 5z + 12 = 0.$$

~~Ex. 4. Find the equation of a sphere for which the circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0, 2x + 3y + 4z = 8$ is a great circle.~~

[Garhwal 81]

Equation of a sphere through given circle is

$$(x^2 + y^2 + z^2 + 7y - 2z + 2) + \lambda (2x + 3y + 4z - 8) = 0.$$

$$\text{Its centre is } \left(-\lambda, -\frac{7+3\lambda}{2}, -\frac{4\lambda-2}{2} \right).$$

If the given circle is a great circle, then the plane $2x + 3y + 4z - 8 = 0$ passes through centre.

$$\therefore -2\lambda - \frac{3}{2} (7 + 3\lambda) - 2 (4\lambda - 2) - 8 = 0.$$

$$\therefore (-2 - \frac{3}{2} - 8) \lambda - 21/2 + 4 - 8 = 0, 29\lambda/2 + 29/2 = 0;$$

$$\therefore \lambda = -1.$$

Putting $\lambda = -1$, we get the equation of sphere as

$$x^2 + y^2 + z^2 - 2x + 4y - 6z + 10 = 0.$$

(b) Obtain the equation of the sphere having the circle

$$x^2 + y^2 + z^2 + 10y - 4z = 8, x - 2y + z = 3$$

as a great circle. [Luck. 82; Madras 78; Utkal 82; Karnataka 83]

$$\text{Ans. } x^2 + y^2 + z^2 + 3x + 4y - z - 17 = 0.$$

Ex. 5. Prove that the plane $x + 2y - z = 4$ cuts the sphere $x^2 + y^2 + z^2 + x + z - 2 = 0$ in a circle of radius unity, and find the equation of sphere which has this circle for one of its great circles.

[Madras 76; Gorakhpur 75; Delhi Hons. 63]

Centre of the sphere is $(\frac{1}{2}, 0, -\frac{1}{2})$ and its radius is

$$R = \sqrt{(\frac{1}{2})^2 + 0 + (\frac{1}{2})^2} = \sqrt{\frac{1}{2}}.$$

If p be the perpendicular from centre of the plane, then

$$p = \frac{\frac{1}{2} + 0 + \frac{1}{2} - 4}{\sqrt{(1+4+1)}} = \frac{3}{\sqrt{6}}.$$

If r be the radius of the circle, then $r^2 = R^2 - p^2$.

$$\text{or } r^2 = \frac{5}{2} - \frac{9}{6} = \frac{15-9}{6} = 1; \therefore r = 1.$$

For the 2nd part proceed as in last question.

Ex 6 (a) Prove that the circles

$$x^2 + y^2 + z^2 - 2x + 3y + 4z - 5 = 0, \quad 5y + 6z + 1 = 0.$$

$$x^2 + y^2 + z^2 - 3x - 4y + 5z - 6 = 0, \quad x + 2y - 7z = 0.$$

lie on the same sphere and find its equation. Also find the value of a for which $x + y + z = a\sqrt{3}$ touches the sphere.

[Madurai 82; Mysore 73; Garhwal 82, 75; Kanpur 78; Meerut 82; Madras 77; Himachal 81; R. Khand 81]

The equation of a sphere through the first circle is

$$(x^2 + y^2 + z^2 - 2x + 3y + 4z - 5) + \lambda (5y + 6z + 1) = 0. \quad \dots(1)$$

Similarly the equation of a sphere through second circle is

$$(x^2 + y^2 + z^2 - 3x - 4y + 5z - 6) + \lambda' (x + 2y - 7z) = 0. \quad \dots(2)$$

If (1) and (2) represent the same sphere, then comparing the coefficients of x, y, z and constant, we get

$$-2 = -3 + \lambda'; \therefore \lambda' = 1.$$

$$3 + 5\lambda = -4 + 2\lambda'; \therefore \lambda = -1.$$

$$\left. \begin{array}{l} 4 + 6\lambda = 5 - 7\lambda' \\ -5 + \lambda = -6 \end{array} \right\} \begin{array}{l} \text{Both are satisfied for values of } \lambda \text{ and} \\ \lambda' \text{ found above.} \end{array}$$

Putting the value of λ in (1) or λ' in (2) we get equation of the sphere as

$$x^2 + y^2 + z^2 - 2x - 2y - 2z - 6 = 0.$$

Its centre is $(1, 1, 1)$. If the given plane touches the sphere, then perpendicular from the centre is equal to radius.

$$\therefore a = (\sqrt{3} \pm 3).$$

(b) Prove that the two circles

$$x^2 + y^2 + z^2 - y + 2z = 0, \quad x - y + z - 2 = 0 \text{ and}$$

$$x^2 + y^2 + z^2 + x - 3y + z - 5 = 0, \quad 2x - y + 4z - 1 = 0$$

lie on the same sphere.

Proceed as above $\lambda = 3, \lambda' = 1$

$$\text{Ans. } x^2 + y^2 + z^2 + 3x - 4y + 5z - 6 = 0.$$

Ex. 7. If r is the radius of the circle,

$$x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0, \quad lx + my + nz = 0,$$

prove that

$$(r^2 + d)(l^2 + m^2 + n^2) = (mw - nv)^2 + (nu - lw)^2 + (lv - mu)^2.$$

[Delhi 75; Rohailkhand 82]

We know that $r^2 = R^2 - p^2$.

$$\begin{aligned} \therefore r^2 &= (u^2 + v^2 + w^2 - d) - \left\{ \frac{(-lu - mv - nw)^2}{\sqrt{(l^2 + m^2 + n^2)}} \right\} \\ \therefore (r^2 + d)(l^2 + m^2 + n^2) &= (l^2 + m^2 + n^2)(u^2 + v^2 + w^2) \\ &\quad - (lu + mv + nw)^2 \text{ etc.} \\ &= (mv - nv)^2 + (nu - lw)^2 + (lv - mu)^2 \end{aligned}$$

by Lagrange's identity (P. 29).

Ex. 8. Find the equation of the circum-circle of triangle whose vertices are $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$. Find also the co-ordinates of its centre and show that its diameter is

$$\sqrt{\left\{ \frac{(b^2 + c^2)(c^2 + a^2)(a^2 + b^2)}{(b^2c^2 + c^2a^2 + a^2b^2)} \right\}}.$$

[Jodhpur 78; Gorakhpur 73; Andhra 75; Meerut 69; Agra 67; Nagpur 62]

Let the given points be A, B, C respectively. The equations of a circle are given by the equations of a sphere through the three points of a circle and a plane.

Equation of the plane ABC is $x/a + y/b + z/c = 1$. .. (1)

We have to find the equation of the sphere through A, B, C . Let the fourth point be taken as origin and the equation of the sphere through O, A, B and C is

$$x^2 + y^2 + z^2 - ax - by - cz = 0 \quad (\text{P. 247})$$

Centre is $(a/2, b/2, c/2)$. Equation of a line through the centre and perpendicular to (1) i.e., parallel to normal are

$$\frac{x - a/2}{1/a} = \frac{y - b/2}{1/b} = \frac{z - c/2}{1/c} = r.$$

Any point on the line is $(r/a + a/2, r/b + b/2, r/c + c/2)$ (3)

If it is the foot of the perpendicular from centre to the plane (1), its co-ordinates will satisfy the equation of the plane.

$$\therefore r \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = 1.$$

$$\therefore r = -\frac{1}{2 \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)}, \text{ substitute in (3),}$$

$$\begin{aligned} \therefore \frac{r}{a} + \frac{a}{2} &= -\frac{1}{2a \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)} + \frac{a}{2} \\ &= \frac{-1 + a^2 \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)}{2a \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)} = \frac{-1 + 1 + a^2 \left(\frac{1}{b^2} + \frac{1}{c^2} \right)}{2a \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right)} \end{aligned}$$